

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
SIMATS ENGINEERING
COMPUTER SCIENCE AND ENGINEERING
CSA17 - ARTIFICIAL INTELLIGENCE

Experiment No. 6: Python Program for the Vacuum Cleaner Problem

Aim

To write a Python program to implement a simple reflex vacuum-cleaner agent that operates in a two-room environment (Room A and Room B), where each room can be either 'Clean' or 'Dirty', and the agent must clean both rooms by sensing and acting based on its current location and the room's status.

Algorithm

1. Start.
2. Represent the environment as a dictionary with two locations, 'A' and 'B', each having a status of either 'Dirty' or 'Clean'.
3. Initialize the agent's starting location (e.g., Location A) and read the initial status of both rooms.
4. Repeat the following steps until both rooms are clean or a maximum number of steps is reached: a) Sense the status of the current location. b) If the current location is 'Dirty', perform the action 'Suck' to clean it (set its status to 'Clean').
5. If the current location is already 'Clean', decide the next action based on simple reflex rules: if at Location A, move Right to Location B; if at Location B, move Left to Location A.
6. After each action, check whether both Room A and Room B are 'Clean'.
7. If both rooms are clean, stop the agent and report the total number of steps taken.
8. If the maximum step limit is reached without both rooms being clean, terminate and report the final state.
9. Print the action taken and the resulting state of the environment at every step.
10. Print the final state of the environment.
11. Stop.

Pseudocode

```
BEGIN
    environment = {A: Dirty, B: Dirty}
    location = A
    steps = 0

    WHILE steps < MAX_STEPS DO
        status = environment[location]

        IF status == Dirty THEN
            ACTION: Suck
            environment[location] = Clean
        ELSE IF location == A THEN
            ACTION: Move Right
            location = B
        ELSE
            ACTION: Move Left
```

```
        location = A
    END IF

    steps = steps + 1

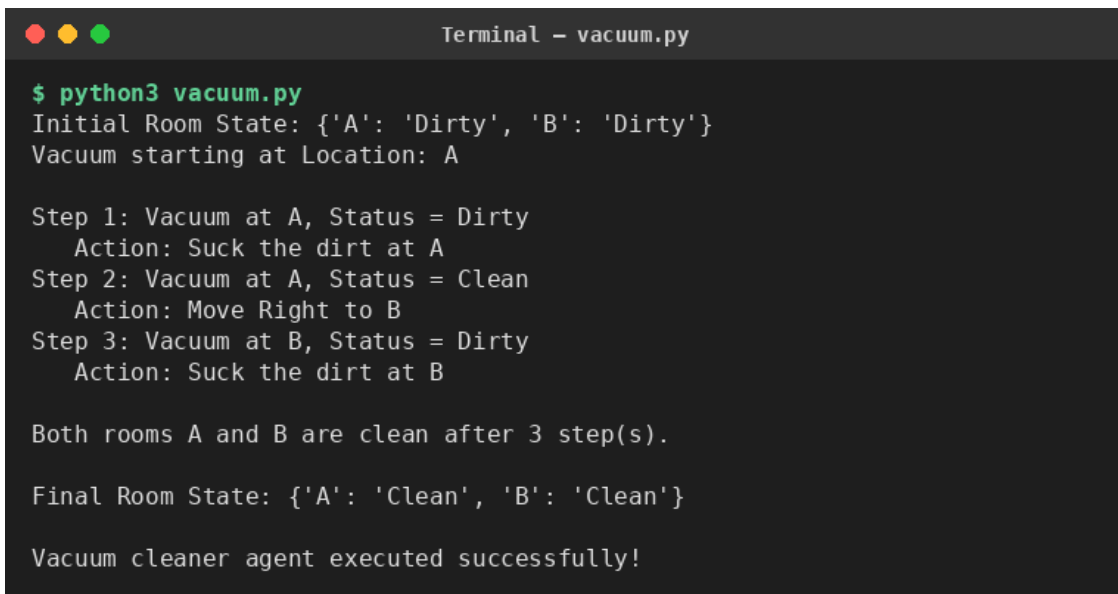
    IF environment[A] == Clean AND environment[B] == Clean THEN
        PRINT "Both rooms clean"
        BREAK
    END IF
END WHILE

PRINT final environment state
END
```

Result

The Python program for the Vacuum Cleaner problem was executed successfully using a simple reflex agent model. Starting with both Room A and Room B in the 'Dirty' state, the agent sensed each location, performed the 'Suck' action to clean dirty rooms, and moved between locations according to the reflex rules. The agent successfully cleaned both rooms within a few steps, and the final environment state showed both Room A and Room B as 'Clean'. Thus, the Vacuum Cleaner problem was solved successfully using Python.

Output Screenshot:

A terminal window titled "Terminal - vacuum.py" with a dark background and light-colored text. It shows the execution of a Python script. The output starts with the initial state of two dirty rooms and the vacuum starting at location A. It then lists three steps: sucking dirt at A, moving to B, and sucking dirt at B. The final output shows both rooms are clean and the vacuum cleaner agent executed successfully.

```
Terminal - vacuum.py

$ python3 vacuum.py
Initial Room State: {'A': 'Dirty', 'B': 'Dirty'}
Vacuum starting at Location: A

Step 1: Vacuum at A, Status = Dirty
Action: Suck the dirt at A
Step 2: Vacuum at A, Status = Clean
Action: Move Right to B
Step 3: Vacuum at B, Status = Dirty
Action: Suck the dirt at B

Both rooms A and B are clean after 3 step(s).

Final Room State: {'A': 'Clean', 'B': 'Clean'}

Vacuum cleaner agent executed successfully!
```